

Christian Sebastian Vargas

(703) 586-3991 | christian.s.vargas4@gmail.com | Virginia, US | linkedin.com/in/christian-s-vargas

EDUCATION

George Mason University

August 2023 – December 2026

Bachelor of Science in Computer Science (GPA: 3.9/4.0)

Fairfax, VA

- Earned Dean's List honors every semester.
- **Selected Courses:** Software Engineering, Data Structures, Analysis of Algorithms, Web Application Development, Mobile Application Development, Introduction to Artificial Intelligence, Data Mining, Deep Learning, Natural Language Processing

TECHNICAL SKILLS

Programming and Web: Java, Python, JavaScript / TypeScript, C, React, Node.js, HTML / CSS, Kotlin, XML, R

Data and AI: Google Agent Development Kit, NumPy, Pandas, scikit-learn, Leaflet, GeoJSON, OpenStreetMap

Databases: MySQL, Google Cloud, Firebase

Tools and APIs: Git / GitHub / GitLab, AWS Lambda, OpenAI, REST APIs

Platforms and Security: Unix / Linux, Alexa Skills Kit (ASK), SHA-256, AES

SELECTED PROJECTS

Google DevFest Hackathon: GridWatch | *HTML / CSS, JS, Google ADK, Python, Firebase* October 2025

- Led the frontend for a 4-person team to ship a real-time city dashboard (map + heatmap, incident list, print-ready report) that helps officials spot outages, traffic congestion, crime, environmental issues and emergencies to send aid.
- Coordinated integration with the AI outputs and public data feeds; simplified the UI so non-technical users can filter by severity and area in one click.
- Collaborated on product direction and organization; documented the UI contract for agent outputs; created mock data so the UI could be built in parallel before backend integration.
- Shipped MVP in 7 hours, placed 2nd overall and invited to present GridWatch to investors / VC firms during DC Startup and Tech Week
- **Website:** <https://gridwatch.dev> | **GitHub:** <https://github.com/christiansvargas/GridWatch>

Spotify Music Recommendation System | *Python, NumPy, Pandas, scikit-learn* August 2025

- Built a music recommendation system coded in Python while employing machine learning techniques such as content-based filtering and cosine similarity to suggest songs from each user's favorites.
- Optimized the system for scalability and responsiveness, implementing a model that can efficiently generate recommendations for large datasets in real time.
- Created online documentation, including clear inline code comments and a README file detailing setup, usage instructions and methodology.
- Maintained a well-structured GitHub repository for version control, fostering ease of collaboration, reproducibility and future scalability.
- **GitHub:** <https://github.com/christiansvargas/recommendation-system>

PatriotHacks: Apollo AI | *OpenAI API, ASK, AWS Lambda, HTML / CSS, JS, React / Node* October 2024

- **Award:** First place in the "Most Likely to Become a Startup" category at PatriotHacks 2024 hackathon
- On a 4-person team, built an end-to-end platform that integrates Azure OpenAI GPT-4o mini with Amazon Alexa for intelligent responses and accessible document summarization.
- Prototyped a 3D-printed Echo Dot casing in CAD; engineered AI prompt / response flows; implemented Git-based change management; integrated the Alexa Skill Management API with the OpenAI API.
- **Devpost/GitHub:** <https://devpost.com/software/apollo-ai-eqvvh3>

PROFESSIONAL EXPERIENCE

Undergraduate Teaching Assistant – Data Structures

January 2025 – June 2025

George Mason University

Fairfax, VA

- Provided academic support for undergraduate students taking Data Structures by answering questions and facilitating discussions on an online message board, escalating complex issues to professors when necessary.
- Assisted during office hours, offering one-on-one guidance and clarifying course concepts such as trees and heaps.
- Proctored exams to ensure academic integrity and adherence to testing procedures; reviewed instructor drafts of programming assignments and provided actionable feedback prior to work dissemination.